S181M116 Derivatives and Alternative Investments: Lecture 3

Swapnil Singh

This lecture supplies the fixed-income inputs used throughout derivative pricing. We study interest-rate conventions, zero rates, discount factors, bond pricing, bootstrapping, forward rates, forward rate agreements, duration, convexity, Treasury bond futures, SOFR futures, and interest rate swaps. The central message is that derivative valuation uses a curve of discount factors, not a single interest rate.

Reading. Hull, Chapters 4, 6, and 7. The practical goal is to move comfortably between quoted rates, discount factors, bond prices, futures prices, and swap values. The mathematical goal is to understand how cash-flow timing and interest-rate sensitivity enter derivative valuation.

Roadmap

Lecture 2 priced forwards using a risk-free rate r. Lecture 3 asks what that input means in real fixed-income markets. A single number is rarely enough: cash flows occur on different dates, quotes use different compounding and day count conventions, and credit and liquidity characteristics differ across rates.

The lecture has three blocks:

1. **Curves.** Convert interest-rate quotes into discount factors, zero rates, forward rates, and bond prices.
2. **Interest-rate futures.** Understand Treasury bond futures, SOFR futures, and duration-based hedging.
3. **Swaps.** Value fixed-for-floating swaps using the same discount curve logic.

Interpretation. The interest rate used in a derivative model is not "the" interest rate in the economy. It is a market-implied discounting input for a particular currency, collateral convention, maturity, and credit/liquidity setting.

Types of Interest Rates

Rates and Basis Points

An interest rate is the promised compensation for lending money over time. There are many rates in the same currency because loans differ in maturity, collateral, credit risk, liquidity, tax treatment, and institutional setting.

Definition 1 (Basis point)
One basis point is one hundredth of one percentage point:

$$1\text{bp} = 0.01\% = 0.0001.$$

A change from 4.25% *to* 4.40% *is a change of* 15 *basis points.*

For derivative pricing, three distinctions are especially important.

Treasury rates Rates implied by government bills and bonds. They are often treated as default-free in nominal terms, but may be affected by regulation, tax, scarcity, and liquidity.

Overnight rates Rates for borrowing and lending over one day. Modern benchmark reform uses overnight rates as the foundation for risk-free reference rates.

Repo rates Secured borrowing rates where cash is lent against collateral, commonly high-quality securities.

Reference Rates

Many contracts set future payments equal to a published reference rate. In the past, LIBOR was the dominant reference rate for many currencies and maturities. The market has moved toward overnight risk-free rates such as SOFR in U.S. dollars, SONIA in sterling, and similar benchmarks in other currencies.

> **LIBOR versus overnight reference rates.** LIBOR was intended to measure unsecured term bank borrowing. New reference rates are built from overnight transactions. This matters because unsecured term bank borrowing includes a credit spread, while overnight reference rates are much closer to risk-free collateralized or short-horizon funding rates.

The move from LIBOR to overnight reference rates also changes timing. A term LIBOR rate was known at the beginning of an interest period. A compounded overnight rate for a period is known only near the end of that period, after the daily overnight rates have been observed.

The Risk-Free Rate in Derivative Pricing

No-arbitrage derivative pricing often constructs a locally or globally riskless position. The return on that position must be consistent with the relevant risk-free discount curve. In practice, derivatives traders usually use market curves built from collateralized overnight rates and swaps rather than simply using Treasury yields.

> **Operational point.** A Treasury yield curve and an overnight-indexed swap discount curve can differ. If the derivative is collateralized and the collateral earns an overnight rate, the collateral convention is part of the pricing problem.

Measuring Interest Rates

Discrete Compounding

Suppose an amount A is invested for n years at a quoted annual rate R. If interest is compounded m times per year, the future value is

$$\mathrm{FV} = A\left(1 + \frac{R}{m}\right)^{mn}.$$

Annual compounding sets $m = 1$, semiannual compounding sets $m = 2$, quarterly compounding sets $m = 4$, and so on.

Example 1 (Compounding frequency)
An investment of 100 at a quoted rate of 10% for one year grows to

$$100(1.10) = 110.00$$

with annual compounding and to

$$100\left(1 + \frac{0.10}{2}\right)^2 = 110.25$$

with semiannual compounding. The quoted rate is the same, but the effective annual return is different.

Two rates with different compounding frequencies are equivalent if they produce the same future value. If R_1 is compounded m_1 times per year and R_2 is compounded m_2 times per year, equivalence requires

$$\left(1 + \frac{R_1}{m_1}\right)^{m_1} = \left(1 + \frac{R_2}{m_2}\right)^{m_2}.$$

Continuous Compounding

As the compounding frequency becomes arbitrarily high, the future value becomes

$$\mathrm{FV} = Ae^{Rn}.$$

Discounting reverses this:

$$\mathrm{PV} = \mathrm{FV}e^{-Rn}.$$

Continuously compounded rates are mathematically convenient because cash-flow discounting, forward-rate calculations, and stochastic process models become cleaner.

Definition 2 (Continuous discount factor)
If $R(0,T)$ is the continuously compounded zero rate from today to maturity T, the discount factor for maturity T is

$$P(0,T) = e^{-R(0,T)T}.$$

The conversion between a rate R_m compounded m times per year and an equivalent continuously compounded rate R_c is

$$R_c = m \ln\left(1 + \frac{R_m}{m}\right),$$

and

$$R_m = m\left(e^{R_c/m} - 1\right).$$

Example 2 (Semiannual to continuous)
A rate of 6% with semiannual compounding has the equivalent continuously compounded rate

$$R_c = 2\ln\left(1 + \frac{0.06}{2}\right) = 2\ln(1.03) \approx 0.0591.$$

The continuous rate is about 5.91%.

Zero Rates and Discount Factors

Zero Rates

A zero-coupon bond pays a single amount at maturity. A zero rate is the rate that discounts that single payment to its current price.

Definition 3 (Zero rate)
The T-year zero rate $R(0,T)$ is the continuously compounded rate that satisfies

$$P(0,T) = e^{-R(0,T)T},$$

where $P(0,T)$ is the current price of receiving one unit of currency at time T.

Equivalently,

$$R(0,T) = -\frac{1}{T}\ln P(0,T).$$

The collection of discount factors $P(0,T)$ or zero rates $R(0,T)$ across maturities is the zero curve.

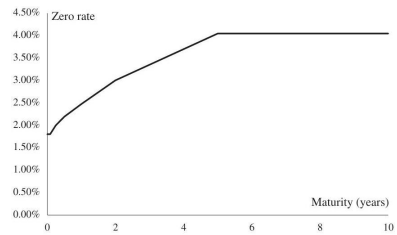

Figure 1: A zero curve gives the discounting input for each maturity. The shape of the curve matters because fixed-income instruments usually contain cash flows on many dates.

Bond Pricing

A coupon bond is a package of dated cash flows. If the bond pays cash flows C_i at dates t_i, its no-arbitrage value is

$$B = \sum_i C_i P(0,t_i).$$

Using continuously compounded zero rates,

$$B = \sum_i C_i e^{-R(0,t_i)t_i}.$$

> **Key distinction.** A bond's yield is a single internal rate of return that prices the entire bond. A zero curve is a term structure of rates that prices each cash flow at its own maturity. Derivative valuation usually needs the zero curve.

Bond Yield

The yield to maturity is the single discount rate y that makes the present value of the bond's promised cash flows equal to its market price:

$$B = \sum_i C_i e^{-yt_i}$$

when y is continuously compounded. The yield is useful as a summary statistic, but it hides the term structure. Two bonds with the same yield can have very different interest-rate sensitivities.

Bootstrapping a Zero Curve

The Bootstrap Logic

Bootstrapping uses traded instruments one maturity at a time. Short-maturity cash instruments pin down early discount factors. Then longer coupon bonds, FRAs, futures, or swaps pin down later discount factors after the earlier ones are already known.

The logic is recursive:

1. Start with the shortest maturity instrument.
2. Use its price to infer the first discount factor.

3. Move to the next maturity.
4. Subtract the present value of earlier cash flows using already known discount factors.
5. Solve for the new discount factor.

Example 3 (A simple bootstrap)
A six-month zero-coupon bond with face value 100 costs 98. Its discount factor is

$$P(0, 0.5) = 0.98.$$

The continuously compounded six-month zero rate is

$$R(0, 0.5) = -\frac{1}{0.5}\ln(0.98) \approx 4.04\%.$$

Now suppose a one-year bond costs 101 and pays 3 in six months and 103 in one year. The one-year discount factor solves

$$101 = 3P(0, 0.5) + 103P(0, 1).$$

Using $P(0, 0.5) = 0.98$,

$$P(0, 1) = \frac{101 - 3(0.98)}{103} \approx 0.9511.$$

The one-year continuously compounded zero rate is

$$R(0, 1) = -\ln(0.9511) \approx 5.01\%.$$

Numerical discipline. Bootstrapping is sensitive to conventions: clean versus dirty prices, accrued interest, day count, coupon frequency, settlement dates, holidays, and interpolation rules. The formula is simple; the market implementation is detailed.

Forward Rates

Forward Rates from Zero Rates

A forward rate is a rate agreed today for borrowing or lending over a future period. Under continuous compounding, the forward rate from T_1 to T_2 is the rate $F(T_1, T_2)$ that makes investing to T_2 directly equivalent to investing to T_1 and then rolling over at the forward rate:

$$e^{R(0,T_2)T_2} = e^{R(0,T_1)T_1} e^{F(T_1,T_2)(T_2-T_1)}.$$

Therefore

$$F(T_1, T_2) = \frac{R(0, T_2)T_2 - R(0, T_1)T_1}{T_2 - T_1}.$$

In discount-factor form,

$$e^{-F(T_1,T_2)(T_2-T_1)} = \frac{P(0,T_2)}{P(0,T_1)}.$$

Example 4 (Forward rate from two zero rates)
The one-year zero rate is 3% and the two-year zero rate is 4%, both with continuous compounding. The one-year forward rate for the second year is

$$F(1,2) = \frac{0.04(2) - 0.03(1)}{2-1} = 0.05.$$

The second-year forward rate is 5%.

Interpretation

Forward rates are break-even rates implied by the current zero curve. They are not generally unbiased forecasts of future spot rates. They include risk premia, convexity effects, liquidity effects, and market segmentation. In pricing, we use them because they are arbitrage-consistent with current market prices.

Forward Rate Agreements

FRA Mechanics

A forward rate agreement (FRA) locks in an interest rate for a future borrowing or lending period. The notional is not exchanged. At the reset date, the market rate for the period is observed and one party compensates the other for the difference between the market rate and the contract rate.

Let:

- N be the notional principal;
- K be the FRA contract rate;
- R be the market rate observed at the reset date;
- τ be the accrual fraction for the underlying loan period.

For the party that receives floating and pays fixed, the payoff paid at the reset date is commonly written as

$$\frac{N\tau(R-K)}{1+R\tau}.$$

The denominator discounts the payoff from the end of the underlying loan period back to the reset date.

Example 5 (FRA settlement)
A trader has an FRA to receive floating and pay fixed on notional EUR 20 million. The contract rate is 4.00%, the realized reference rate is 4.60%, and the accrual period is three months, so $\tau = 0.25$. The settlement at the reset date is

$$\frac{20{,}000{,}000(0.25)(0.046 - 0.040)}{1 + 0.046(0.25)} \approx 29{,}659.$$

The payoff is positive because the trader locked in paying 4.00% and the market rate moved to 4.60%.

Duration

Macaulay Duration

Duration measures the weighted average timing of a bond's cash flows and the first-order sensitivity of bond value to yield changes. With continuous compounding, if a bond price is

$$B = \sum_i C_i e^{-yt_i},$$

then Macaulay duration is

$$D = \frac{\sum_i t_i C_i e^{-yt_i}}{B}.$$

Differentiating the bond price with respect to y gives

$$\frac{\Delta B}{B} \approx -D\Delta y.$$

Longer-duration bonds lose more value when yields rise and gain more value when yields fall.

Example 6 (Duration of a two-cash-flow bond)
A bond pays 5 in one year and 105 in two years. The continuously compounded yield is 4%. Its price is

$$B = 5e^{-0.04} + 105e^{-0.08} \approx 101.78.$$

Its duration is

$$D = \frac{1(5e^{-0.04}) + 2(105e^{-0.08})}{101.78} \approx 1.95.$$

For a 20 basis point parallel yield increase, the duration approximation gives

$$\frac{\Delta B}{B} \approx -1.95(0.002) = -0.0039,$$

or a price decline of about 0.39%.

Modified Duration

When yields are quoted with periodic compounding, the price sensitivity is usually stated using modified duration. If Macaulay duration is D and the yield is compounded m times per year, then

$$D^* = \frac{D}{1 + y/m}.$$

The approximation becomes

$$\frac{\Delta B}{B} \approx -D^* \Delta y.$$

Portfolio Duration

The duration of a portfolio is the value-weighted average of the durations of its components:

$$D_P = \sum_j w_j D_j,$$

where w_j is the fraction of portfolio value invested in instrument j. This lets a manager estimate interest-rate exposure at the portfolio level.

Convexity

Why Duration Is Not Enough

Duration is a linear approximation. Bond prices are curved functions of yields: as yields fall, price gains accelerate; as yields rise, price losses are smaller than a purely linear approximation would predict for sufficiently large moves.

With continuous compounding, convexity is

$$C = \frac{\sum_i t_i^2 C_i e^{-y t_i}}{B}.$$

The duration-convexity approximation is

$$\frac{\Delta B}{B} \approx -D\Delta y + \frac{1}{2} C (\Delta y)^2.$$

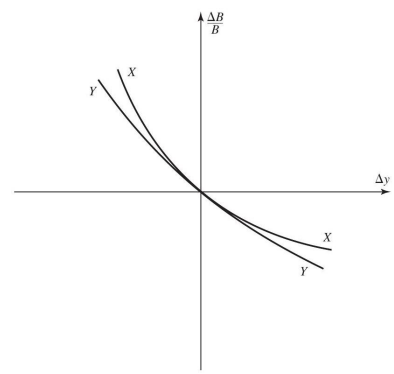

Figure 2: Duration is the tangent-line approximation. Convexity captures the curvature of the bond price-yield relation.

Example 7 (Adding convexity)
Suppose a bond has duration 6.8 and convexity 55. If yields increase by 50 basis points, $\Delta y = 0.005$. *The duration-only approximation is*

$$-6.8(0.005) = -3.40\%.$$

The convexity correction is

$$\frac{1}{2}(55)(0.005)^2 = 0.0006875,$$

or 0.06875%. The duration-convexity approximation is therefore

$$-3.40\% + 0.06875\% = -3.33125\%.$$

Term Structure Theories

The zero curve contains market information, but its shape can be interpreted in several ways.

Expectations theory Forward rates reflect expected future short rates.

Market segmentation Different investors and borrowers concentrate on different maturities, so supply and demand by maturity affects yields.

Liquidity preference Investors may require compensation for holding longer-maturity bonds, creating a term premium.

For derivative pricing, the most important point is not choosing one theory. It is using a curve that is internally consistent with market prices and then being clear about whether a forward rate is a pricing rate or a forecast.

Interest Rate Futures

Day Count and Quotation Conventions

Fixed-income markets are convention-heavy. Interest accruals depend on the day count convention. Common examples include actual/actual, actual/360, and 30/360. The same quoted annual rate can imply different cash interest over a calendar period if the day count differs.

> **Do not ignore conventions in implementation.** In classroom derivations we often simplify by assuming exact year fractions. In market pricing, small differences in day count, settlement, accrued interest, and holiday calendars can change the quoted price and the hedge.

Treasury Bill and Bond Quotes

Treasury bills are discount instruments and are often quoted using bank discount rates. Treasury bonds are coupon instruments and are usually quoted as clean prices. The invoice amount paid by a buyer is the dirty price:

$$\text{dirty price} = \text{clean price} + \text{accrued interest}.$$

The distinction matters for futures delivery and bond pricing because actual cash paid at settlement includes accrued interest.

Treasury Bond Futures

The Delivery Option

Treasury bond futures allow the short futures position to choose from a basket of deliverable bonds. Because different deliverable bonds have different coupons and maturities, the exchange uses conversion factors to standardize delivery.

Definition 4 (Conversion factor)
The conversion factor adjusts the invoice price of a deliverable bond in a Treasury bond futures contract. Ignoring accrued interest, the invoice price is approximately

$$\textit{futures quote} \times \mathrm{CF}.$$

The full invoice amount is

$$\text{invoice amount} = \text{futures settlement price} \times \mathrm{CF} + \text{accrued interest}.$$

Cheapest-to-Deliver

The short chooses the bond that is cheapest to deliver. Ignoring accrued interest and delivery timing details, the short compares

$$\text{cash bond price} - (\text{futures price}) \times \mathrm{CF}.$$

The bond with the lowest cost is the cheapest-to-deliver bond.

Example 8 (Cheapest-to-deliver comparison)
The futures price is 112. Three deliverable bonds have clean prices and conversion factors:

Bond	*Cash price*	CF
A	118.20	1.0500
B	124.40	1.1030
C	110.10	0.9800

Ignoring accrued interest, the delivery costs are

$$\begin{aligned} A &: 118.20 - 112(1.0500) = 0.60, \\ B &: 124.40 - 112(1.1030) = 0.864, \\ C &: 110.10 - 112(0.9800) = 0.34. \end{aligned}$$

Bond C is cheapest to deliver.

Futures Price from a Known CTD Bond

If the cheapest-to-deliver bond is known and delivery options are ignored, the futures price can be approximated from cost of carry:

$$F_0 \approx \frac{(B_0 - I)e^{rT}}{\mathrm{CF}},$$

where B_0 is the current dirty price of the CTD bond, I is the present value of coupons received before delivery, r is the financing rate, T is time to delivery, and CF is the conversion factor.

> **Connection to Lecture 2.** Treasury bond futures use the same cash-and-carry logic as forwards on investment assets. The added complications are coupons, accrued interest, conversion factors, and delivery options.

SOFR Futures

Reference Rate Futures

Short-term interest-rate futures settle on reference rates. Older textbooks and market histories discuss Eurodollar futures, which were linked to three-month LIBOR. The post-LIBOR U.S. market uses SOFR futures, based on the secured overnight financing rate.

A standard interest-rate futures quote is often written as

$$\text{futures quote} = 100 - \text{annualized rate}.$$

Thus a quote of 95.25 corresponds to an implied rate of 4.75%.

Example 9 (SOFR futures quote)
A SOFR futures quote is 96.80. The implied annualized rate is

$$100 - 96.80 = 3.20\%.$$

If the quote falls to 96.65, the implied rate rises to 3.35%. A falling price therefore corresponds to rising interest-rate expectations.

Compounding and Convexity

SOFR for a longer period is built from overnight rates compounded over the period. The annualized compounded rate has the form

$$\left[\prod_{i=1}^{n}\left(1 + r_i \frac{d_i}{360}\right) - 1\right]\frac{360}{D},$$

where r_i is the annualized overnight rate for day i, d_i is the number of calendar days covered by that rate, and $D = \sum_i d_i$.

Futures rates and forward rates are not always identical because futures are settled daily. Daily settlement creates a convexity effect when the futures payoff is correlated with interest rates. For short maturities the adjustment may be small; for longer maturities and volatile rates it can matter.

Duration-Based Hedging with Futures

Hedging a Bond Portfolio

A bond portfolio loses value when yields rise. A manager can hedge by shorting interest-rate futures. If rates rise, the bond portfolio loses value but the short futures position gains.

For a portfolio value P with duration D_P, and a futures contract with price F and duration D_F for the asset underlying the futures contract, the hedge ratio is approximately

$$N^* = \frac{PD_P}{FD_F}.$$

If the portfolio yield and the futures-implied yield do not move one-for-one, include a yield beta:

$$N^* = \frac{PD_P}{FD_F}\frac{\Delta y_P}{\Delta y_F}.$$

Example 10 (Duration hedge)
A bond portfolio is worth EUR 40 million and has duration 5.5. The futures contract price is EUR 120,000 and the duration of the deliverable bond exposure is 7.2. Assuming parallel yield moves,

$$N^* = \frac{40{,}000{,}000(5.5)}{120{,}000(7.2)} \approx 254.6.$$

The manager shorts about 255 futures contracts.

Limits of Duration Hedges

Duration hedging is approximate. It assumes small yield changes, stable duration, parallel shifts, and a reliable relationship between the portfolio yield and the futures yield. The hedge can fail when the yield curve twists, credit spreads move, the cheapest-to-deliver bond changes, or convexity is important.

Interest Rate Swaps

Plain Vanilla Swap Mechanics

A plain vanilla interest rate swap exchanges fixed interest payments for floating interest payments on a notional principal. The notional is

used to calculate interest but is not exchanged.

In a fixed-for-floating swap:

- one party pays a fixed rate S on notional N;
- the other party pays a floating reference rate, such as compounded SOFR plus or minus a spread;
- payments are netted on each payment date.

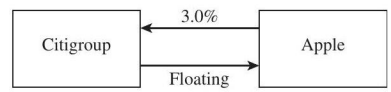

Figure 3: A fixed-for-floating swap exchanges fixed-rate payments for floating rate payments on the same notional.

> **Economic use.** A borrower with floating-rate debt can enter a swap to pay fixed and receive floating. The floating receipts offset the floating debt payments, transforming the liability into a fixed-rate liability.

Net Payments

Suppose payments occur every τ years. If the fixed rate is S and the floating rate set for the period is L, the net payment from the fixed payer to the floating payer is

$$N\tau(S - L).$$

If $L > S$, the fixed payer receives the difference. If $S > L$, the fixed payer pays the difference.

The Swap Rate

Par Swap Rate

At initiation, a standard swap is usually priced so that its value is zero. The fixed rate that makes the value zero is the par swap rate.

Let payment dates be $T_1, \ldots, T_n$ with accrual fractions $\alpha_1, \ldots, \alpha_n$. Let $P(0, T_i)$ be discount factors. For a standard par swap with no exchange of principal, the fixed rate is

$$S = \frac{1 - P(0, T_n)}{\sum_{i=1}^{n} \alpha_i P(0, T_i)}.$$

The denominator is the fixed-leg annuity.

Example 11 (Par swap rate)
A two-year swap pays annually. Discount factors are $P(0,1) = 0.96$ and $P(0,2) = 0.915$. With $\alpha_1 = \alpha_2 = 1$, the par fixed rate is

$$S = \frac{1 - 0.915}{0.96 + 0.915} = 0.0453.$$

The par swap rate is about 4.53%.

Relationship to the Zero Curve

The swap rate is a weighted average of forward rates, where the weights are discounted accrual factors. When the zero curve slopes upward, longer swap rates usually exceed shorter rates. But the swap rate is not simply the zero rate for the same maturity.

Valuing Interest Rate Swaps

Bond Decomposition

A fixed-for-floating interest rate swap can be valued as the difference between a floating-rate bond and a fixed-rate bond.

For a party that receives floating and pays fixed,

$$V_{\text{swap}} = B_{\text{float}} - B_{\text{fixed}}.$$

For a party that receives fixed and pays floating,

$$V_{\text{swap}} = B_{\text{fixed}} - B_{\text{float}}.$$

The fixed-rate bond value is

$$B_{\text{fixed}} = NS \sum_{i=1}^{n} \alpha_i P(0, T_i) + NP(0, T_n).$$

The floating-rate bond value is close to par immediately after a reset date. Between reset dates, it equals the present value of the next known floating payment plus principal.

Example 12 (Swap value after rates change)
A EUR 10 million swap has one and two years remaining, annual payments, and fixed rate 4.00%. The current discount factors are $P(0,1) = 0.955$ *and* $P(0,2) = 0.900$. *For the party receiving fixed and paying floating, the fixed bond value is*

$$\begin{aligned} B_{fixed} &= 10{,}000{,}000(0.04)(0.955 + 0.900) \\ &\quad + 10{,}000{,}000(0.900) \\ &= 9{,}742{,}000. \end{aligned}$$

If the swap is just after a reset date, approximate $B_{float} = 10{,}000{,}000$. *Then*

$$V_{receive\ fixed} = 9{,}742{,}000 - 10{,}000{,}000 = -258{,}000.$$

The receive-fixed position is negative because current rates have moved above the old fixed rate.

FRA Decomposition

The same swap can be viewed as a portfolio of forward rate agreements. For a receive-floating, pay-fixed swap, the value is approximately

$$V = N \sum_i \alpha_i (F_i - S) P(0, T_i),$$

where F_i is the forward rate for payment period i. This representation makes clear that a swap locks in a sequence of future short-rate exchanges.

Using Swaps to Transform Assets and Liabilities

Transforming a Liability

Suppose a firm has floating-rate debt and wants fixed-rate debt. It can enter a swap where it pays fixed and receives floating. The floating leg received on the swap offsets the floating payments on the debt. The firm is left paying the fixed swap rate plus any borrowing spread on its debt.

If the firm instead has fixed-rate debt and wants floating-rate exposure, it can enter the opposite swap: receive fixed and pay floating.

Transforming an Asset

A bank that owns a fixed-rate loan can use a swap to receive floating and pay fixed, thereby transforming a fixed-rate asset into a floating-rate asset. This can help manage net interest income when liabilities reprice more quickly than assets.

> **Risk-management point.** A swap changes interest-rate exposure, but it does not eliminate all risk. Basis risk, credit risk, collateral liquidity, model risk, and operational risk remain.

Currency Swaps

Fixed-for-Fixed Currency Swaps

In a currency swap, the parties exchange cash flows in different currencies. Unlike a plain vanilla interest rate swap, the principals are usually exchanged at the start and re-exchanged at maturity.

For example, one party might pay fixed dollars and receive fixed euros. The currency swap value is the domestic value of one bond minus the domestic value of the other bond. If S_0 is the domestic

price of one unit of foreign currency, then a receive-foreign, pay-domestic position has value

$$V = S_0 B_{\text{foreign}} - B_{\text{domestic}}.$$

Uses

Currency swaps can transform borrowing from one currency into another, exploit market access differences, and manage long-term foreign-currency cash-flow exposure. They introduce exchange-rate exposure and cross-currency funding spreads, so valuation requires both interest-rate curves and the spot exchange rate.

Credit Risk and Other Swaps

Credit Risk in Swaps

A swap's credit exposure changes through time. At initiation, a standard swap has value near zero, but later rate movements can make it positive to one party and negative to the other. The party for whom the swap has positive value is exposed to counterparty default.

Collateral agreements, central clearing, netting, and margin reduce this risk, but they do not make it disappear. This is why modern derivative valuation often distinguishes between the trade's promised cash flows and adjustments for funding, collateral, capital, and counterparty credit risk.

Other Swap Structures

The same swap logic extends beyond standard interest rate swaps:

- basis swaps exchange one floating rate for another;
- overnight indexed swaps exchange fixed payments for compounded overnight floating payments;
- equity swaps exchange equity-index returns for fixed or floating interest;
- commodity swaps exchange a fixed commodity price for a floating spot or index price;
- volatility swaps and variance swaps exchange payments linked to realized volatility or variance.

The valuation details differ, but the common structure is the same: identify the future cash flows, choose the relevant state variables and discount curve, and value the contract by no-arbitrage or a pricing model.

Worked Problems

Example 13 (Discount factor from zero rate)
The continuously compounded two-year zero rate is 3.75%. The discount factor is

$$P(0,2) = e^{-0.0375(2)} = e^{-0.075} \approx 0.9277.$$

Receiving EUR 1 million in two years is worth approximately

$$1{,}000{,}000(0.9277) = 927{,}700.$$

Example 14 (Bond price from zero curve)
A bond pays 4 in one year and 104 in two years. The one-year zero rate is 3% and the two-year zero rate is 3.8%, both continuously compounded. The price is

$$B = 4e^{-0.03(1)} + 104e^{-0.038(2)} \approx 3.88 + 96.38 = 100.26.$$

Example 15 (Forward rate from discount factors)
Suppose $P(0,1) = 0.9704$ *and* $P(0,1.5) = 0.9418$. *The continuously compounded forward rate from year 1 to year 1.5 solves*

$$e^{-F(0.5)} = \frac{0.9418}{0.9704}.$$

Thus

$$F = -\frac{1}{0.5}\ln\left(\frac{0.9418}{0.9704}\right) \approx 6.0\%.$$

Example 16 (Treasury bond futures approximation)
The dirty price of the CTD bond is 128.50. Coupons with present value 2.10 will be received before delivery. The financing rate is 4%, time to delivery is 0.5 years, and the conversion factor is 1.18. The approximate futures price is

$$F_0 = \frac{(128.50 - 2.10)e^{0.04(0.5)}}{1.18} = \frac{126.40e^{0.02}}{1.18} \approx 109.31.$$

Example 17 (Swap rate from discount factors)
A three-year swap pays annually. Discount factors are

$$P(0,1) = 0.970, \quad P(0,2) = 0.930, \quad P(0,3) = 0.885.$$

The par swap rate is

$$S = \frac{1 - 0.885}{0.970 + 0.930 + 0.885} = \frac{0.115}{2.785} \approx 4.13\%.$$

Checklist for Students

After this lecture you should be able to:

- explain why derivative pricing uses discount curves rather than one universal interest rate;

- convert rates across compounding frequencies;
- compute continuously compounded discount factors;
- define zero rates, par yields, and forward rates;
- price a coupon bond from a zero curve;
- bootstrap a simple zero curve from bond prices;
- calculate forward rates from zero rates or discount factors;
- explain the settlement logic of a forward rate agreement;
- compute duration and use it for first-order interest-rate risk;
- add convexity to improve the price-change approximation;
- distinguish clean price, dirty price, and accrued interest;
- explain conversion factors and cheapest-to-deliver logic in Treasury bond futures;
- interpret SOFR futures quotes;
- construct a duration-based futures hedge for a bond portfolio;
- describe fixed-for-floating swap cash flows;
- derive the par swap rate from discount factors;
- value a swap as the difference between a fixed-rate bond and a floating-rate bond;
- explain how swaps transform assets and liabilities;
- identify the main risks that remain after entering an interest-rate swap.

Practice Questions

1. Convert a rate of 8% compounded quarterly into an equivalent continuously compounded rate.
2. Convert a continuously compounded rate of 5% into an equivalent semiannually compounded rate.
3. A three-year zero rate is 4.2% with continuous compounding. Compute the three-year discount factor.
4. A bond pays 6 in one year and 106 in two years. The one-year and two-year zero rates are 3% and 4%. Compute the bond price.

5. A six-month zero-coupon bond costs 97.80 per 100 face value. Compute the six-month continuously compounded zero rate.

6. A one-year coupon bond costs 101.20 and pays 4 in six months and 104 in one year. Use the discount factor from the previous question to bootstrap the one-year discount factor.

7. The one-year zero rate is 3.5% and the two-year zero rate is 4.1%, both continuously compounded. Compute the one-year forward rate for year 2.

8. A trader receives floating and pays fixed in an FRA. The notional is 50 million, the contract rate is 3.7%, the realized rate is 4.1%, and the accrual fraction is 0.25. Compute the settlement at the reset date.

9. A bond pays 3 in one year and 103 in two years. Its continuously compounded yield is 4.5%. Compute its price and duration.

10. A bond has duration 8 and convexity 80. Approximate the percentage price change when yields rise by 25 basis points.

11. Explain why a high-convexity bond is more valuable than a low-convexity bond with the same price and duration, holding other risks fixed.

12. A Treasury bond futures price is 118. One deliverable bond has price 126 and conversion factor 1.060. Another has price 121 and conversion factor 1.020. Which is cheaper to deliver, ignoring accrued interest?

13. The dirty price of the CTD bond is 115. Coupons with present value 1.5 are paid before delivery, the financing rate is 3%, delivery is in four months, and the conversion factor is 0.92. Approximate the futures price.

14. A SOFR futures quote falls from 95.40 to 95.10. What happens to the implied rate?

15. A bond portfolio is worth USD 25 million and has duration 6. A futures contract has price USD 110,000 and duration 7.5. Compute the approximate number of contracts for a duration hedge.

16. Draw the cash flows for a party that pays fixed and receives floating in an interest rate swap.

17. Annual discount factors for years 1, 2, and 3 are 0.965, 0.925, and 0.880. Compute the three-year annual-pay par swap rate.

18. A receive-fixed swap has fixed rate 5%, notional 20 million, and two annual payments remaining. Current discount factors are 0.96 and 0.91. If the floating-rate bond value is approximately par, compute the swap value.

19. Explain why a swap can have zero value at initiation but large credit exposure later.

20. Give one example each of an interest rate swap, a basis swap, a currency swap, and a commodity swap.